CISC 473: Deep Learning — Midterm Report

P22: Cross-Architecture Knowledge Distillation

Group Members:

Lee Xuan 20630324 (Role: Project Lead)

Fong Yee Xin 20629927 (Role: Engineering Lead)

GitHub Repository: https://github.com/l33zard/cakd.git

Date: October 22, 2026

Abstract

Knowledge distillation (KD) compresses a large teacher network into a compact student, but most feature-level KD methods assume teacher and student share an architectural family. We study the harder cross-architecture case of a convolutional ResNet-50 teacher and a DeiT-Tiny vision-transformer student on CIFAR-100, where spatial feature maps must be related to patch-token sequences. To date we have implemented a deterministic CIFAR-100 pipeline (45k/5k/10k train/validation/test split), CIFAR-adapted ResNet-50 and DeiT-Tiny models, a shared training and evaluation engine, and resumable training scripts for the teacher and for the from-scratch student baseline. After 100 epochs the ResNet-50 teacher reaches 75.35% top-1 and 93.47% top-5 test accuracy (single seed), below the reference result discussed in the proposal under a longer training schedule. The from-scratch DeiT-Tiny student has completed an initial four-epoch preliminary run (25.24% validation top-1), confirming that the supervised student pipeline runs end to end; its completed run will establish the lower bound that distillation must improve on. The next stages are vanilla KD, layer-wise CKA measurement, and the proposed projected-feature alignment loss.

Introduction and Problem Statement

Edge deployment favours small models, and vision transformers are increasingly the architecture of choice for compact students, yet the strongest teachers available for small-image tasks are often CNNs. Logit-only KD [1,3] transfers across architectures but discards the teacher's intermediate knowledge, while feature-level methods [2,4,5,7] were designed for CNN-to-CNN pairs whose feature maps are directly comparable. A CNN produces a $C \times H \times W$ spatial map and a ViT produces an $N \times D$ token sequence, so feature transfer between them needs an explicit alignment step.

Research question. *Does aligning projected intermediate CNN features with ViT patch tokens, using a CKA-based loss, give a DeiT-Tiny student accuracy gains on CIFAR-100 beyond logit-only KD, at unchanged inference cost, and is any gain accompanied by higher teacher–student CKA similarity?*

Scope changes since the proposal. Following instructor feedback, we narrowed the project to one cross-architecture pair: ResNet-50 teacher → DeiT-Tiny student on CIFAR-100. The required progression is the data pipeline, teacher, supervised student, vanilla KD, CKA measurement,

projected CKA alignment, and focused ablations. CRD, ReviewKD, DKD, TinyImageNet, the ResNet-110 → MobileNetV2 pair, and attention-map alignment are out of scope for this project.

Related Work (Updated)

Logit-based distillation. Hinton et al. [1] train the student to match the teacher's temperature-softened outputs, exposing the "dark knowledge" in non-target class probabilities. DKD [3] shows that the classical loss couples non-target knowledge to the teacher's target confidence and decouples the two terms, matching feature-based methods at logit-only cost. Because they act only on output distributions, both transfer unchanged across architectures, which makes vanilla KD our base objective and DKD the strongest logit-only reference.

Feature-based distillation. FitNets [5] introduced intermediate "hints" with a learnable regressor to reconcile channel widths; our projection module generalises this regressor from a channel mismatch to a map-versus-token mismatch. CRD [2] maximises a contrastive bound on teacher–student mutual information over pooled embeddings, ReviewKD [4] lets deeper student stages attend to shallower teacher stages, and RKD [7] transfers pairwise distance and angle structure, which is insensitive to representation dimensionality. All were developed and evaluated almost entirely on CNN pairs.

Cross-architecture distillation. DeiT [9] trains a ViT student from a CNN teacher (RegNetY) through a dedicated distillation token and finds that a convolutional teacher outperforms a transformer teacher, suggesting the CNN's inductive bias is itself useful knowledge; however, it distils logits only and modifies the student architecture with an extra token. Liu et al. [11] project CNN features into a transformer-like space with partially cross-attentive and group-wise linear projectors, and OFA-KD [12] projects intermediate features of heterogeneous students into the logit space and distils there, reporting gains on CIFAR-100 and ImageNet across CNN, ViT and MLP pairs. Our approach differs in aligning token-level representations directly with a CKA objective [6], adding no tokens to the student, and using CKA as a diagnostic of where transfer succeeds. These papers provide context for heterogeneous distillation, but they are not implemented as baselines in the narrowed project.

Method	Year	Key idea	Relation to our work
Vanilla KD [1]	2015	Match temperature-softened logits	Base loss term; first baseline
FitNets [5]	2015	L_2 hint loss with learnable regressor	Ancestor of our projection module
CKA [6]	2019	Invariant similarity of representations	Diagnostic and alignment loss
CRD [2]	2020	Contrastive teacher–student MI bound	Feature-level baseline (CNN pairs)
ReviewKD [4]	2021	Cross-stage review of teacher features	Motivates layer-pair ablation
DeiT [9]	2021	Distillation token, CNN teacher → ViT	Logit-only CNN→ViT reference
DKD [3]	2022	Decoupled target / non-target KD	Strongest logit-only baseline
OFA-KD [12]	2023	Distil heterogeneous features in logit space	Closest cross-architecture competitor

Methodology

Model Architecture

Teacher. ResNet-50 [8] from `torchvision`, randomly initialized (no pre-trained weights). For 32×32 inputs the 7×7 stride-2 stem is replaced by a 3×3 stride-1 convolution, the initial max-pool is removed, and the classifier outputs 100 classes. The `layer3` output is a $1024 \times 8 \times 8$ map. The implementation contains 23,705,252 trainable parameters.

Student. DeiT-Tiny [9] from `timm`, randomly initialised, with the standard 192-dimensional embedding, 12 blocks and 3 heads, but with input size 32 and patch size 4. This gives an 8×8 grid of 64 patch tokens plus a class token (sequence 65×192). The implementation contains 5,380,132 trainable parameters.

Planned alignment module (not yet implemented). The primary pair is ResNet layer3 → DeiT transformer block 8. The teacher's 8×8 `layer3` map is reshaped from $B \times 1024 \times 8 \times 8$ to $B \times 64 \times 1024$ and passed through a learnable linear projection $1024 \to 192$. The student representation uses the 64 patch tokens after block 8; the CLS token is excluded. Both projected representations are reshaped to $(B \cdot 64) \times 192$, treating patch/token positions across the minibatch as observations and embedding coordinates as features. We use centered linear CKA,

$$\mathrm{CKA}(X, Y) = \frac{\|X^\top Y\|_F^2}{\|X^\top X\|_F \, \|Y^\top Y\|_F + \varepsilon}, \qquad \mathcal{L}_{\text{align}} = 1 - \mathrm{CKA}(X, Y).$$

The total proposed student objective is

$$\mathcal{L} = (1 - \alpha)\, \mathcal{L}_{\text{CE}} + \alpha\, \tau^2\, \mathrm{KL}(\sigma(z_T/\tau) \,\|\, \sigma(z_S/\tau)) + \beta \left(1 - \mathrm{CKA}(P(F_T),\, F_S)\right),$$

where P is the projection and F_S contains patch tokens only. The projection is discarded after training, so inference cost is unchanged. The focused layer-pair ablation compares early layer2 → block4, middle layer3 → block8, and late layer4 → block12. These pairs are not all shape-compatible before adaptation. For the early pair, the teacher output has shape $B \times 512 \times 16 \times 16$;

it is bilinearly resized to 8×8, flattened to $B \times 64 \times 512$, and projected with a separate $512 \to 192$ linear layer. The middle pair naturally produces $B \times 64 \times 1024$ before its $1024 \to 192$ projection. For the late pair, the $B \times 2048 \times 4 \times 4$ teacher output is bilinearly resized to 8×8, flattened to $B \times 64 \times 2048$, and projected with a separate $2048 \to 192$ linear layer. After these operations, every teacher pair produces $B \times 64 \times 192$, matching the student's 64 patch tokens; both are then reshaped to $(B \cdot 64) \times 192$ for CKA.

Training Setup

Hyperparameter	Teacher (ResNet-50)	Student (DeiT-Tiny, scratch)
Optimiser	SGD, momentum 0.9	AdamW
Learning rate	0.1, cosine annealing	3×10^{-4}, cosine decay (no warm-up in the final protocol)
Weight decay	5×10^{-4}	0.05
Batch size	128	128
Epochs	100	100
Loss function	Cross-entropy	Cross-entropy
Data augmentation	Random crop (pad 4), horizontal flip	Same
Random seeds used	42 initially; 123 and 999 planned	42 initially; 123 and 999 planned

The checkpoint with the best validation top-1 accuracy is kept and evaluated once on the test set. The trained teacher checkpoint (`teacher_best.pt`) and metric logs are available at https://drive.google.com/drive/folders/1J2HeM9IKQTrtBiroXC0C_Mx8FZVZoYu9?usp=sharing; see the repository README for usage. Training scripts save the best validation checkpoint and epoch-level CSV metrics. Both scripts support resuming from a checkpoint, restoring the model, optimiser, and learning-rate schedule state; smoke-test outputs are used only for pipeline verification and are not presented as research results.

Dataset and Preprocessing

CIFAR-100 [10]: 60,000 32×32 colour images in 100 balanced classes. The official 50,000-image training set is split deterministically (split seed 42) into 45,000 training and 5,000 validation images; the official 10,000-image test set is used only for final evaluation. Images are normalised with per-channel CIFAR-100 mean and standard deviation; training images additionally receive a random 32×32 crop after 4-pixel padding and a random horizontal flip, while validation and test images are not augmented. Classes are balanced, so no re-weighting is applied.

Evaluation protocol. Top-1 and top-5 accuracy on the test set, reported as mean $\pm$ std over 3 seeds, together with parameter count and training time.

Preliminary Results

Baseline Reproduction

Target: Match the published number within 1–2% before proceeding to your proposed modification.

Model	Top-1 (%)	Top-5 (%)	Params
ResNet-50 teacher, ours (100 ep., test)	75.35	93.47	23,705,252
DeiT-Tiny student, scratch (4/100 ep., val.)	25.24†	57.12†	5,380,132

Table 1: Preliminary results on CIFAR-100. Our results are from a single seed (42); the remaining two seeds are scheduled for Weeks 7–8, after which we will report mean ± std. †Validation accuracy after 4 of 100 epochs, not a final result. These are preliminary single-seed results and must be rerun or accompanied by their saved logs before the final report.

Qualitative Analysis

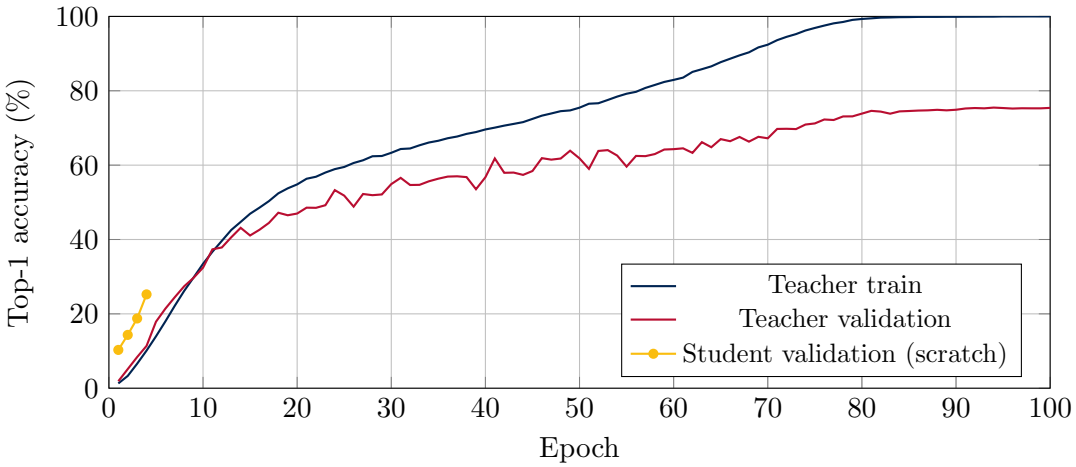


Figure 1: Top-1 accuracy per epoch on CIFAR-100. The ResNet-50 teacher's training accuracy reaches ≈100% while validation accuracy plateaus at ≈75% (best 75.48% at epoch 94), showing a large generalisation gap. The DeiT-Tiny curve shows only the initial four-epoch preliminary run.

Discussion

Teacher reproduction. Our teacher reaches 75.35% test top-1, about 4 points below the reference value cited in the proposal, so it does not yet meet the original reproduction target. The most likely causes are the shorter 100-epoch budget and differences between published and local training recipes. A teacher that is substantially stronger than the student remains useful for the planned distillation comparison, but the teacher gap must be reported when interpreting KD gains.

Overfitting. Training accuracy reaches 99.99% and training loss falls to 0.007, while validation accuracy plateaus near 75% from about epoch 85 onwards (Fig. 1). The teacher has effectively memorised the 45k-image training set with only crop and flip augmentation. Stronger regularisation (e.g. mixup or longer schedules) could narrow this gap, though we prioritise the distillation experiments over further tuning the teacher.

Student. After four epochs the student reaches 25.24% validation top-1. This is an incomplete preliminary run, so no conclusion about final student accuracy or distillation gains can be drawn yet. Vision transformers trained from scratch on small datasets are known to underperform CNNs because they lack a locality prior [9]; the size of the final teacher–student gap is the headroom that distillation must recover.

Challenges and Adaptations

Compute interruptions. Training uses the Google Colab free tier, whose sessions end after roughly 12 hours or on inactivity, deleting anything stored on the runtime's local disk. We moved checkpoints and metric logs to Google Drive and added a `--resume` option that restores model, optimiser and learning-rate schedule from the last best checkpoint, which let the 100-epoch teacher run complete across several sessions.

Correct resumption. An early version restarted the cosine schedule from the wrong epoch after a resume and mixed zero- and one-based epoch numbers; we fixed both so a resumed run follows the same learning-rate curve as an uninterrupted one. Because only the best checkpoint is saved, resuming can repeat a few epochs and append duplicate rows to the metrics log; these are removed before plotting.

Feature-shape compatibility. Choosing patch size 4 for DeiT-Tiny gives an 8×8 token grid with 64 patch tokens. The primary layer3 $\rightarrow$ block8 pair already has this spatial resolution, so it needs no interpolation. The early layer2 output is 16×16 with 256 spatial positions, while the late layer4 output is 4×4 with 16 positions; neither can be compared directly with the student's 64 tokens. Both ablation maps are therefore bilinearly resized to 8×8 and use their own channel projection before flattening to $(B \cdot 64) \times 192$. We verify the raw model shapes with temporary forward hooks in `src/models.py`.

Timeline. We are behind the proposal schedule, which placed vanilla KD in Week 3, CRD in Week 4, DKD in Week 5 and ReviewKD and CKA in Week 6. So far the data pipeline, both models, the shared training engine, the complete 100-epoch teacher run and the first 4 epochs of the scratch student are done. The 100-epoch teacher alone consumed several free-tier GPU sessions, and the student run paused when the Colab quota was exhausted. We are moving remaining training to Kaggle (30 GPU-hours per week) in addition to Colab and re-ordering the plan below so the core CNN$\rightarrow$ViT comparison is completed first.

Updated Week 7–10 Plan

Week	Tasks	Responsible
7	Finish scratch student (100 epochs); implement vanilla KD; run temperature and loss-weight screening; implement layer-wise CKA tooling	All
8	Implement projected-CKA alignment; run the specified layer-pair and loss-weight ablations	All
9	Run selected configurations with seeds 42, 123, and 999; produce CKA comparisons and efficiency analysis	All
10	Final report, presentation, single-command reproduction and README **(Final due)**	All

Preliminary Novel Contribution (Optional Bonus)

The planned contribution is a reproducible CKA-based study of CNN-to-ViT distillation. We will compare the primary layer pair and the early/middle/late ablation pairs, then test whether higher CKA similarity is associated with higher classification accuracy. This contribution remains prospective until the alignment implementation and experiments are complete.

AI Tool Disclosure

AI assistance was used to inspect the repository, suggest report structure and wording, and help identify implementation and documentation issues. The group remains responsible for the code, experiments, results, interpretation, and final verification of this report.

Contribution Declaration

Member	Contributions (this period)	%
Lee Xuan	Project coordination; literature review; scope refinement; teacher training and evaluation; preliminary-results analysis; report and presentation preparation	50%
Fong Yee Xin	CIFAR-100 data pipeline; ResNet-50 and DeiT-Tiny implementation; shared training/evaluation engine; student and teacher scripts; checkpointing and reproducibility support	50%

References

[1] Hinton, G., Vinyals, O., and Dean, J., 2015. "Distilling the Knowledge in a Neural Network." *NeurIPS Deep Learning and Representation Learning Workshop*. arXiv:1503.02531.

[2] Tian, Y., Krishnan, D., and Isola, P., 2020. "Contrastive Representation Distillation." *ICLR*. arXiv:1910.10699.

[3] Zhao, B., Cui, Q., Song, R., Qiu, Y., and Liang, J., 2022. "Decoupled Knowledge Distillation." *CVPR*. arXiv:2203.08679.

[4] Chen, P., Liu, S., Zhao, H., and Jia, J., 2021. "Distilling Knowledge via Knowledge Review." *CVPR*. arXiv:2104.09044.

[5] Romero, A., Ballas, N., Kahou, S. E., Chassang, A., Gatta, C., and Bengio, Y., 2015. "FitNets: Hints for Thin Deep Nets." *ICLR*. arXiv:1412.6550.

[6] Kornblith, S., Norouzi, M., Lee, H., and Hinton, G., 2019. "Similarity of Neural Network Representations Revisited." *ICML*. arXiv:1905.00414.

[7] Park, W., Kim, D., Lu, Y., and Cho, M., 2019. "Relational Knowledge Distillation." *CVPR*. arXiv:1904.05068.

[8] He, K., Zhang, X., Ren, S., and Sun, J., 2016. "Deep Residual Learning for Image Recognition." *CVPR*. arXiv:1512.03385.

[9] Touvron, H., Cord, M., Douze, M., Massa, F., Sablayrolles, A., and Jégou, H., 2021. "Training Data-Efficient Image Transformers & Distillation through Attention." *ICML*. arXiv:2012.12877.

[10] Krizhevsky, A., 2009. "Learning Multiple Layers of Features from Tiny Images." Technical report, University of Toronto.

[11] Liu, Y., Cao, J., Li, B., Hu, W., Ding, J., and Li, L., 2022. "Cross-Architecture Knowledge Distillation." *ACCV*. arXiv:2207.05273.

[12] Hao, Z., Guo, J., Han, K., Tang, Y., Hu, H., Wang, Y., and Xu, C., 2023. "One-for-All: Bridge the Gap Between Heterogeneous Architectures in Knowledge Distillation." *NeurIPS*. arXiv:2310.19444.